

7-Day Node.js Learning Roadmap

Day 1 – Node.js Fundamentals

Goal: Understand what Node.js is and how it runs JavaScript outside the browser.

Topics:

- What is Node.js
- How Node.js works (Event Loop)
- Installing Node.js
- Running JavaScript using Node
- Built-in modules: fs, path, os

Practice:

Create a file reader using the fs module and print system info using os.

Outcome:

You understand how Node executes JavaScript.

Day 2 – Modules & NPM

Goal: Learn how Node projects are structured.

Topics:

- CommonJS modules
- require()
- module.exports
- Node Package Manager (NPM)
- package.json
- Installing packages

Practice:

Create a simple math module and import it into another file.

Outcome:

You can create reusable modules.

Day 3 – HTTP Server

Goal: Build your first backend server.

Topics:

- Node http module
- Request & Response
- Basic routing

Practice:

Create routes for:

/

/about

/contact

Outcome:

You understand backend server basics.

Day 4 – Express.js

Goal: Use Express to simplify server creation.

Topics:

- Installing Express
- Creating routes
- Middleware
- Request parameters
- Query parameters

Practice:

Build API routes:

GET /users

GET /users/:id

POST /users

Outcome:

You can create REST APIs.

Day 5 – Database Integration (MongoDB)

Goal: Store and retrieve data.

Topics:

- MongoDB basics
- Mongoose

- Database connection
- Schema & Models
- CRUD operations

Practice:

Create APIs to:

Create user

Get users

Delete user

Outcome:

You can connect Node to a database.

Day 6 – Authentication

Goal: Secure APIs with authentication.

Topics:

- JWT authentication
- Password hashing
- Login & Register APIs

Authentication Flow:

Register → Hash password → Save user

Login → Verify password → Generate JWT token

Outcome:

You can create secure APIs.

Day 7 – Build a Mini Project

Project Idea: Blog API

Features:

- Register user
- Login user
- Create post
- Update post
- Delete post
- Get posts

Tech Stack:

Node.js, Express, MongoDB, JWT

Optional Deployment Platforms:

Render, Vercel, Railway

Suggested Daily Study Plan

Activity	Time
Learning	1 hour
Coding	2 hours
Practice	1 hour

After 7 Days You Should Know

- Node.js fundamentals
- Express framework
- REST API development
- Database integration
- Authentication
- Backend project structure